

$$x(n) - 2x(n-1) = n \cdot 2^n u(n) + \delta(n)$$

$$\mathcal{Z}\{x(n-1)\} = \frac{x(z)}{z} \rightarrow (1)$$

$$\mathcal{Z}\{\delta(n)\} = 1 \rightarrow (2)$$

$$\mathcal{Z}\{a^n u(n)\} = \frac{1}{1 - \frac{a}{z}}$$

using

$$\mathcal{Z}\{n \cdot f(n)\} = -z \frac{d}{dz} f(z)$$

$$\mathcal{Z}\{n a^n u(n)\} = \frac{a/z}{(1 - a/z)^2} \rightarrow (3)$$

Applying z-transform, using (1), (2), (3)

$$x(z) - 2 \frac{x(z)}{z} = \frac{z/z}{(1 - z/z)^2} + 1$$

$$x(z) \left(1 - \frac{2}{z}\right) = 1 + \frac{z/z}{(1 - z/z)^2}$$

$$x(z) = \frac{1}{\left(1 - \frac{2}{z}\right)} + \frac{z/z}{\left(1 - \frac{2}{z}\right)^3}$$

Applying inverse

$$x(n) = 2^n u(n) + \left(\frac{n}{2}\right) 2^n u(n) \frac{n(n+1)}{2}$$

$$x(n) = 2^n u(n) + \frac{n(n+1)}{2} 2^n u(n)$$

$$x(n) = \left(1 + \frac{n}{2} + \frac{n^2}{2}\right) 2^n u(n)$$

for $n \geq 0$

$$u(n) = 1$$

$$T(n) = 2^n \left(1 + \frac{n}{2} + \frac{n^2}{2}\right)$$